

Assignment Number: 1.5

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BATCH: 24

Task 1: AI-Generated Logic Without Modularization (String Reversal Without Functions)

❖ Scenario

You are developing a basic text-processing utility for a messaging application.

❖ Task Description

Use GitHub Copilot to generate a Python program that:

- Reverses a given string
- Accepts user input
- Implements the logic directly in the main code
- Does not use any user-defined functions

❖ Expected Output

- Correct reversed string
- Screenshots showing Copilot-generated code suggestions
- Sample inputs and outputs

```
1 #write a python program to reverse a string without using functions
2 def reverse_string(s):
3     reversed_str = ""
4     for i in range(len(s) - 1, -1, -1):
5         reversed_str += s[i]
6     return reversed_str
7
8 input_string = "Hello World"
9 output_string = reverse_string(input_string)
10 print("Original String:", input_string)
11 print("Reversed String:", output_string)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS + v ?

```
Reversed String: dlrow olleH
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> & C:/Users/yadav/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/yadav/OneDrive/Desktop/AI-LAB/assign.py
Original String: Hello World
Reversed String: dlrow olleH
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB> |
```

Task 2: Efficiency & Logic Optimization (Readability Improvement)

❖ Scenario

The code will be reviewed by other developers.

❖ Task Description

Examine the Copilot-generated code from Task 1 and improve it by:

- Removing unnecessary variables
- Simplifying loop or indexing logic
- Improving readability
- Use Copilot prompts like:
 - “Simplify this string reversal code”
 - “Improve readability and efficiency”

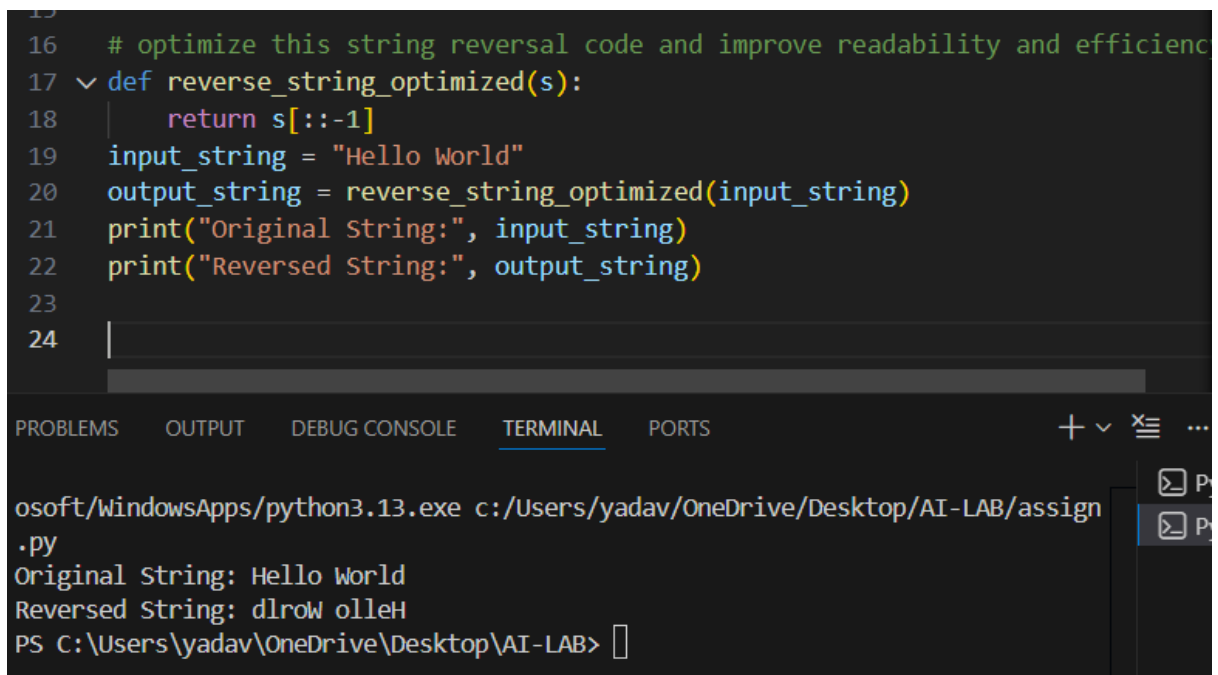
Hint:

Prompt Copilot with phrases like

“optimize this code”, “simplify logic”, or “make it more readable”

❖ Expected Output

- Original and optimized code versions
- Explanation of how the improvements reduce time complexity



```
16 # optimize this string reversal code and improve readability and efficiency
17 def reverse_string_optimized(s):
18     return s[::-1]
19 input_string = "Hello World"
20 output_string = reverse_string_optimized(input_string)
21 print("Original String:", input_string)
22 print("Reversed String:", output_string)
23
24
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

osoft/WindowsApps/python3.13.exe c:/Users/yadav/OneDrive/Desktop/AI-LAB/assign
.py
Original String: Hello World
Reversed String: dlrow olleH
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB>

Task 3: Modular Design Using AI Assistance (String Reversal Using Functions)

❖ Scenario

The string reversal logic is needed in multiple parts of an application.

❖ Task Description

Use GitHub Copilot to generate a function-based Python program that:

- Uses a user-defined function to reverse a string
- Returns the reversed string
- Includes meaningful comments (AI-assisted)

❖ Expected Output

- Correct function-based implementation
- Screenshots documenting Copilot's function generation
- Sample test cases and outputs

```
26 #give string reversal code with using functions
27 def reverse_string(s):
28     return s[::-1]
29 input_string = "Hello World"
30 output_string = reverse_string(input_string)
31 print("Original String:", input_string)
32 print("Reversed String:", output_string)
33
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Original String: Hello World
Reversed String: dlrow olleH
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB>

Task 5: AI-Generated Iterative vs Recursive Fibonacci Approaches (Different Algorithmic Approaches to String Reversal)

❖ Scenario

Your mentor wants to evaluate how AI handles alternative logic paths.

❖ Task Description

Prompt GitHub Copilot to generate:

- A loop-based string reversal approach
- A built-in / slicing-based string reversal approach
- ❖ Expected Output
 - Two correct implementations
 - Comparison discussing:
 - Execution flow
 - Time complexity
 - Performance for large inputs
 - When each approach is appropriate

```
36 #give different approaches to reverse a string like a loop based and built
37 def reverse_string_loop(s):
38     reversed_str = ""
39     for i in range(len(s) - 1, -1, -1):
40         reversed_str += s[i]
41     return reversed_str
42 def reverse_string_slicing(s):
43     return s[::-1]
44 input_string = "Hello World"
45 output_string_loop = reverse_string_loop(input_string)
46 output_string_slicing = reverse_string_slicing(input_string)
47 print("Original String:", input_string)
48 print("Reversed String using loop:", output_string_loop)
49 print("Reversed String using slicing:", output_string_slicing)
50 '''
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Original String: Hello World
Reversed String using loop: dlrow olleH
Reversed String using slicing: dlrow olleH
PS C:\Users\yadav\OneDrive\Desktop\AI-LAB>